



2015 MacBook Pro recalled

Apple voluntarily announces recall of all 2015 Macbook Pro batteries due to risk of overheating. <https://dgit.in/july19-64>



Walmart blocking burglary

Walmart is using AI powered cameras to prevent theft at the check out counters. <https://dgit.in/july19-65>



Taking the heat

A bizarre request sends Agent001 onto a wild-goose chase for SSD heat sinks

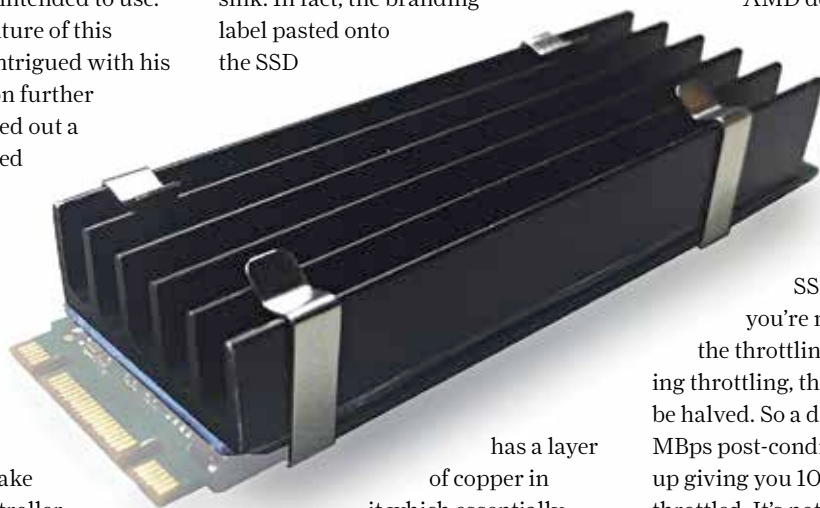
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bout a month ago, a friend of mine asked me to grab an SSD heatsink for him on my way back from a

junket. He was extremely specific about getting a heatsink that covers the entire surface and should come with a backplate that can screw on so that he could vary the clamping pressure for whatever thermal interface material he intended to use. Given the peculiar nature of this request, I was quite intrigued with his thought process. Upon further questioning, he pointed out a video which mentioned that it's better for SSDs if the NAND is kept at a higher temperature but the SSD controller has to be kept cool to prevent thermal throttling. What he basically wanted to do was to take the heat from the controller of the SSD, which runs quite hot, and channel it to the NAND via the heatsink. One can understand going to such extensive lengths in the data centre space where every fraction of performance matters but for a consumer drive, this seemed a bit too much. So, I decided to investigate a little further into this.

NVMe SSDs are usually installed in the M.2 slot that sits right underneath the first x16 PCIe slot. This also means that this particular slot will always have hot exhaust air from the GPU blowing directly onto it. After running the GPU at peak load as well as normal load, I noticed that the SSD throttles rather quickly when the GPU is running at full load. Note that this is a plain SSD with no attached heat-sink. In fact, the branding label pasted onto the SSD



has a layer of copper in it which essentially makes it a heatsink of sorts. The surprising part? It was that the SSD would throttle around the 10-15 second mark if the GPU was running a power-virus. If that were to be scaled down, and the temperatures similar to running popular AAA titles were to be simulated, then the throttling happened much later. It happened ap-

proximately, a minute into a consistent write operation.

It seems, in most day to day situations, the SSD is not going to be running a persistent write for this long to even get throttled in the first place. You're not going to be copying a movie while running a video rendering workload and be playing some video game in the foreground. Wait ... that's pretty much what Intel and

AMD demonstrate when they

want to showcase multi-tasking. Perhaps, you might be experiencing throttling after all. It's just that with the ridiculous speeds that NVMe

SSDs have these days, you're not going to perceive the throttling because even during throttling, the speeds are going to be halved. So a drive that does 2800 MBps post-conditioning might end up giving you 1000-1400 MBps when throttled. It's not like you'll know the difference unless you are staring at a progress bar.

So we've figured that the SSD (without a proper heatsink) will throttle during a persistent write cycle while the ambient temperature is high. The time it throttles at is a function of the heat generated by the SSD controller (which ramps up quick when actively